Probability & Distributions – Exam-Oriented Study Notes

These notes are **strictly exam-focused** and compiled from past year questions, ISM concept slides, practice problems, and standard theory. Content is ordered **module-wise**, with **key formulas, exam tips, and solved examples**.

MODULE 1: BASIC STATISTICS & INTRODUCTION TO PROBABILITY

1. Measures of Central Tendency

Definitions - **Mean**: Arithmetic average - **Median**: Middle value after sorting - **Mode**: Most frequent value

Important Properties - Mean is **affected by outliers** - Median is **robust to outliers** → preferred for skewed data - Mode is useful for **categorical data**

Empirical Relation (Moderately skewed data) Mode = 3 × Median − 2 × Mean

Exam Tip - If data is **heavily skewed**, always justify why **median** is better than mean

Example Data: 50, 52, 55, 58, 150 - Mean = 73 (distorted by outlier) - Median = 55 (better representative)

YouTube (English) - Khan Academy – Mean, Median, Mode: https://www.youtube.com/watch?v=7Hk9jct2ozY - StatQuest – Mean vs Median: https://www.youtube.com/watch?v=itQEwESWDKg

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2. Measures of Variability

Range Range = Max − Min

Variance & Standard Deviation - Population variance: divide by **N** - Sample variance: divide by **n − 1**

Why n − 1? (Bessel's Correction) - Using sample mean reduces one degree of freedom - Dividing by n underestimates variance

Formulas - Sample Variance: $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$ - Standard Deviation: $s = \sqrt{s^2}$

Exam Tip - If the word **sample** appears → always use **n − 1**

YouTube - StatQuest – Why divide by n−1: https://www.youtube.com/watch?v=DXM8l6JzSxE - Khan Academy – Variance & SD: https://www.youtube.com/watch?v=VJjv8HqCj1A

3. Skewness, Five-Point Summary & Outliers

Skewness - Right skewed: Mean > Median > Mode - Left skewed: Mean < Median < Mode - Symmetric: Mean = Median = Mode

Five-Number Summary - Minimum, Q1, Median, Q3, Maximum

Interquartile Range (IQR) IQR = Q3 − Q1

Outlier Detection - Lower bound: Q1 − 1.5 × IQR - Upper bound: Q3 + 1.5 × IQR

Exam Tip - Always show **sorted data** before finding quartiles

YouTube - StatQuest – Boxplots & IQR: https://www.youtube.com/watch?v=fHLhBnmwUM0

MODULE 2: PROBABILITY THEORY

4. Basic Probability Rules

Addition Rule P(A ∪ B) = P(A) + P(B) − P(A ∩ B)

Mutually Exclusive Events P(A ∩ B) = 0

Independent Events P(A ∩ B) = P(A) × P(B)

Very Important Exam Statement - Mutually exclusive events (with non-zero probability) are **NOT independent**

YouTube - StatQuest – Independence vs Mutual Exclusivity: https://www.youtube.com/watch?v=2rId6lYjK7Q

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5. Conditional Probability

Formula P(A | B) = P(A ∩ B) / P(B)

Typical Exam Problems - Cards and balls (without replacement) - Dice and coins

YouTube - Khan Academy – Conditional Probability: https://www.youtube.com/watch?v=KqzM_kN2KxE

MODULE 3: BAYES' THEOREM

6. Bayes' Theorem

Formula $P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$

Law of Total Probability $P(B) = \Sigma P(B \mid A_i) P(A_i)$

Common Exam Applications - Medical testing - Factory defect problems - Weather & machine accuracy

Naive Bayes (Concept) - Assumes independence between features - Used in text classification

Exam Tip - Always calculate **P(B)** first using total probability

YouTube - StatQuest – Bayes Theorem (Best): https://www.youtube.com/watch?v=HZGCoVF3YvM - Khan Academy – Bayes Intuition: https://www.youtube.com/watch?v=9wz6o7pZ5pE

MODULE 4: RANDOM VARIABLES

7. Random Variables

Discrete Random Variable - Uses PMF - $\Sigma P(X = x) = 1$

Continuous Random Variable - Uses PDF - $P(X = x) = 0$ - Probabilities found using **area under curve**

YouTube - StatQuest – Random Variables: https://www.youtube.com/watch?v=9YJZ3Z3zYpQ

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8. Expectation & Variance

Rules - $E(aX + b) = aE(X) + b$ - $Var(aX + b) = a^2 Var(X)$

Exam Tip - Expectation is linear (no independence needed)

MODULE 5: DISCRETE PROBABILITY DISTRIBUTIONS

9. Bernoulli Distribution

- Single trial
- Mean = p
- Variance = $p(1 - p)$

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10. Binomial Distribution

Conditions - Fixed number of trials (n) - Independent trials - Constant probability (p)

PMF $P(X = k) = C(n, k) p^k (1 - p)^{n-k}$

Mean & Variance - Mean = np - Variance = np(1 − p)

YouTube - StatQuest – Binomial Distribution: https://www.youtube.com/watch?v=8idr1WZ1A7Q

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11. Poisson Distribution

When to Use - Rare events - Large n, small p - Given rate λ

PMF $P(X = k) = (\lambda^k e^{-l}) / k!$

Mean = Variance = λ

YouTube - Khan Academy – Poisson Distribution: https://www.youtube.com/watch?v=5Kd3b5Zc3bM

MODULE 6: CONTINUOUS DISTRIBUTIONS

12. Normal Distribution

Properties - Bell-shaped, symmetric - Mean = Median = Mode

Empirical Rule - 68% → μ ± 1σ - 95% → μ ± 2σ - 99.7% → μ ± 3σ

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13. Z-Score

Formula $z = (x - \mu) / \sigma$

Applications - Probability calculation - Percentiles - Comparing relative performance

Exam Tip - Always sketch or imagine the normal curve

YouTube - StatQuest – Z-Scores: https://www.youtube.com/watch?v=itj4lC2sXgQ - Khan Academy – Normal Distribution: https://www.youtube.com/watch?v=rzFX5NWojp0

FINAL EXAM CHECKLIST

✔️ Mean vs Median decision ✔️n vs n−1 (Bessel's correction) ✔️Mutually exclusive ≠ independent ✔️ Bayes theorem with total probability ✔️Binomial vs Poisson conditions ✔️Z-score & table usage

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End of Exam-Oriented Study Notes